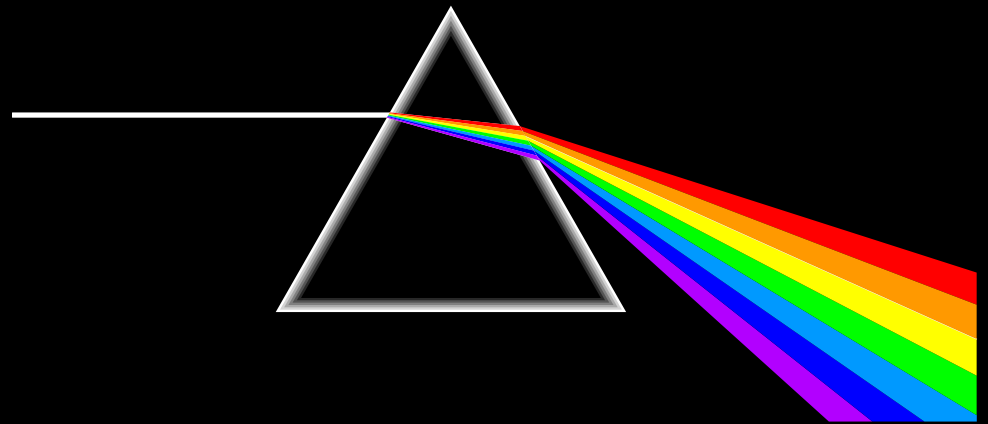


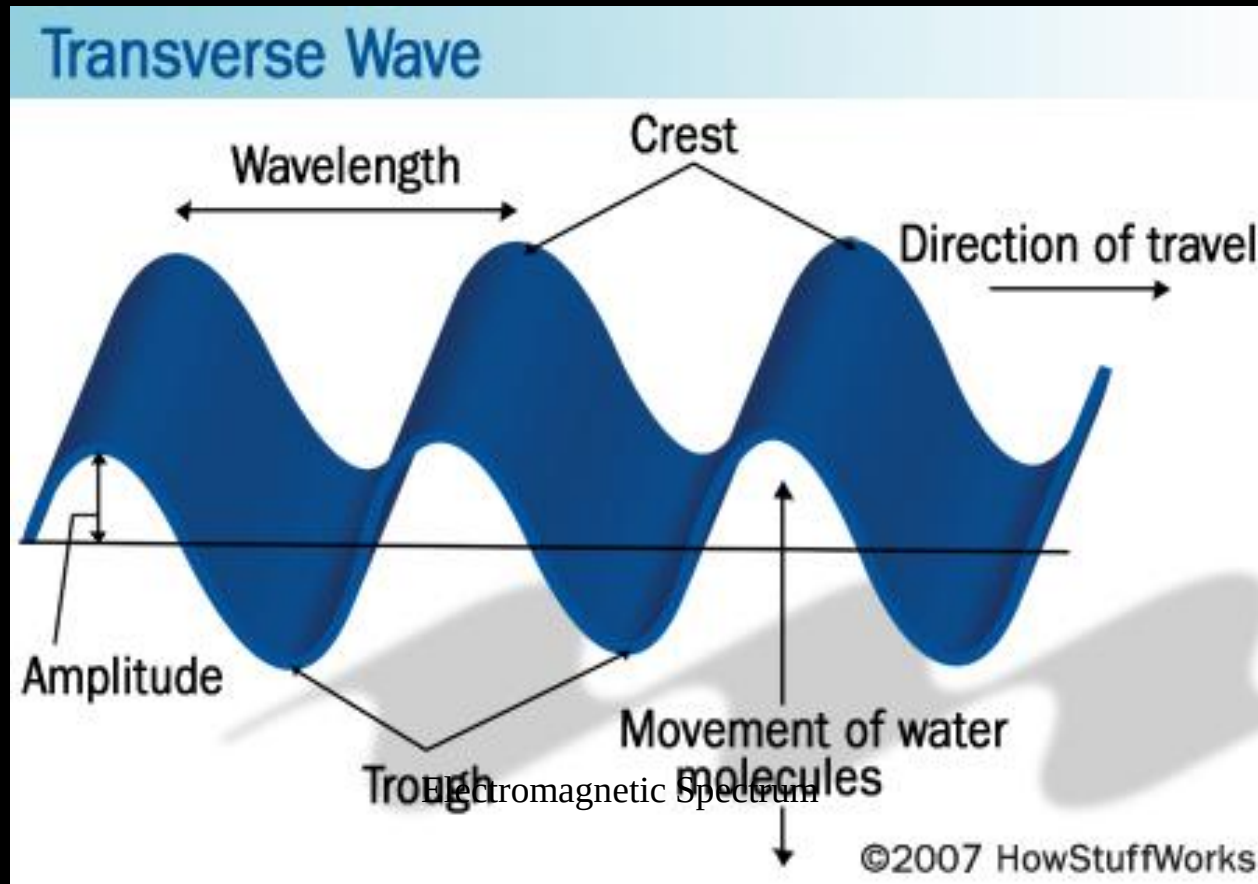
Electromagnetic Waves & the Electromagnetic Spectrum

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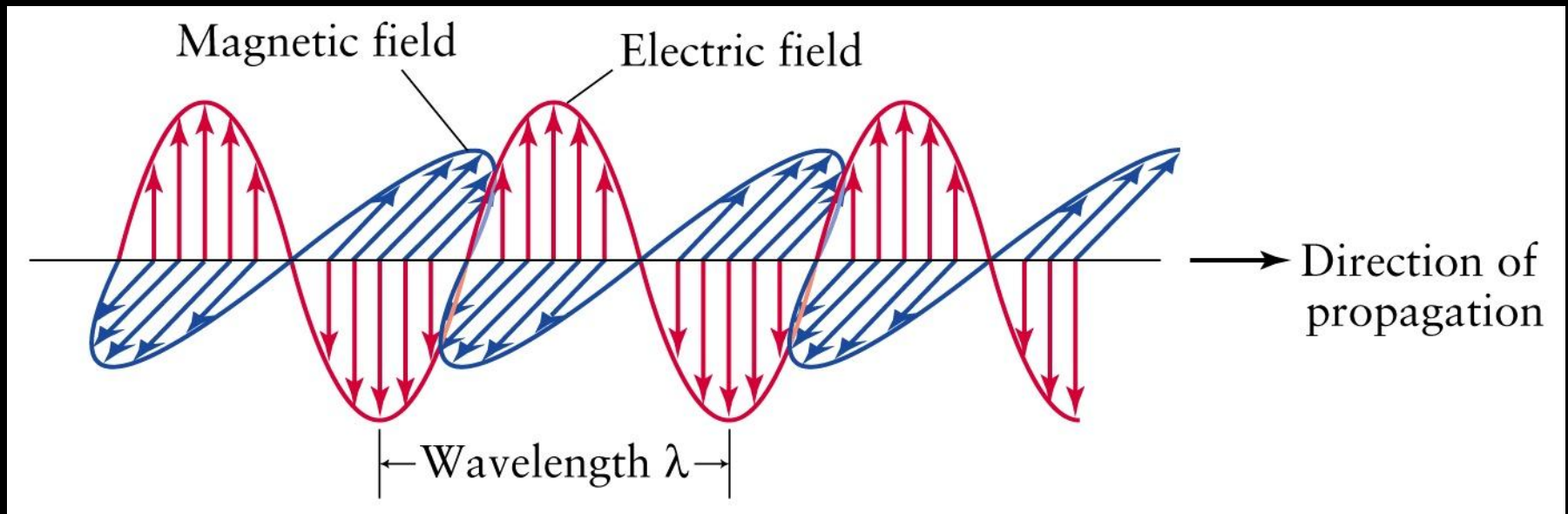
Electromagnetic Waves

- Transverse waves without a medium!
- (They can travel through empty space)



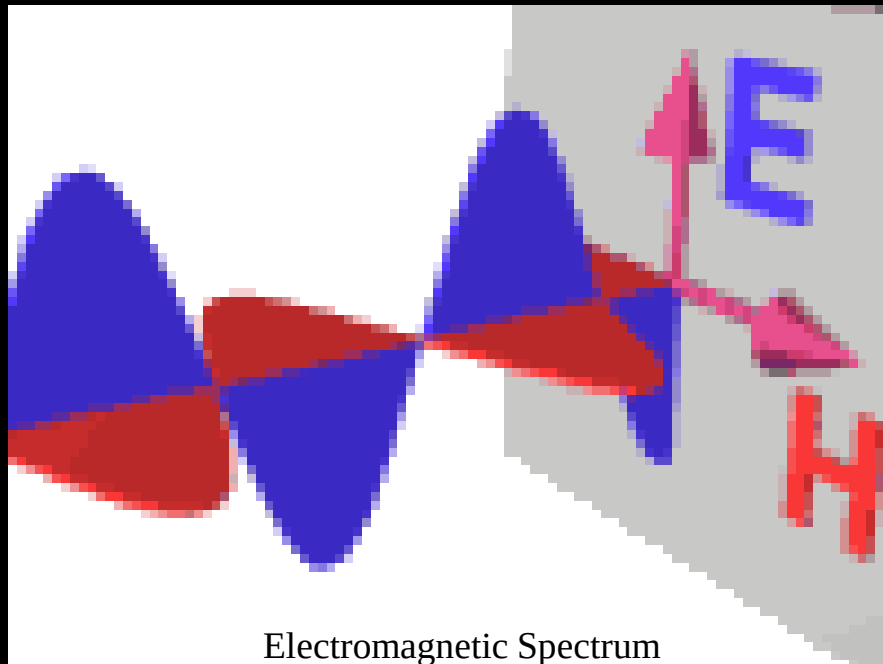
- They travel as vibrations in electrical and magnetic fields.

- Have some magnetic and some electrical properties to them.



- When an electric field changes, so does the magnetic field. The changing magnetic field causes the electric field to change. When one field vibrates—so does the other.

RESULT-An electromagnetic wave.



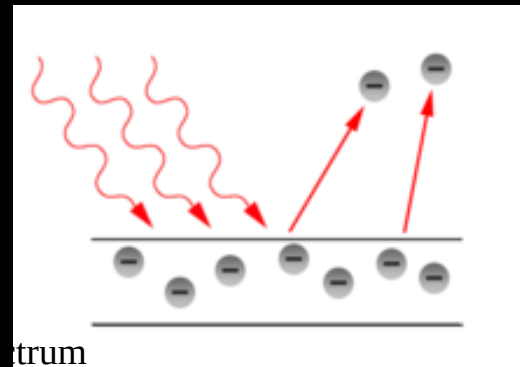
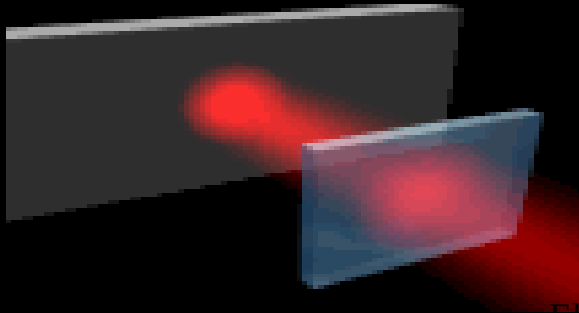
Electromagnetic waves travel VERY FAST - around 300,000 kilometres per second (the speed of light).

At this speed they can go around the world 8 times in one second.

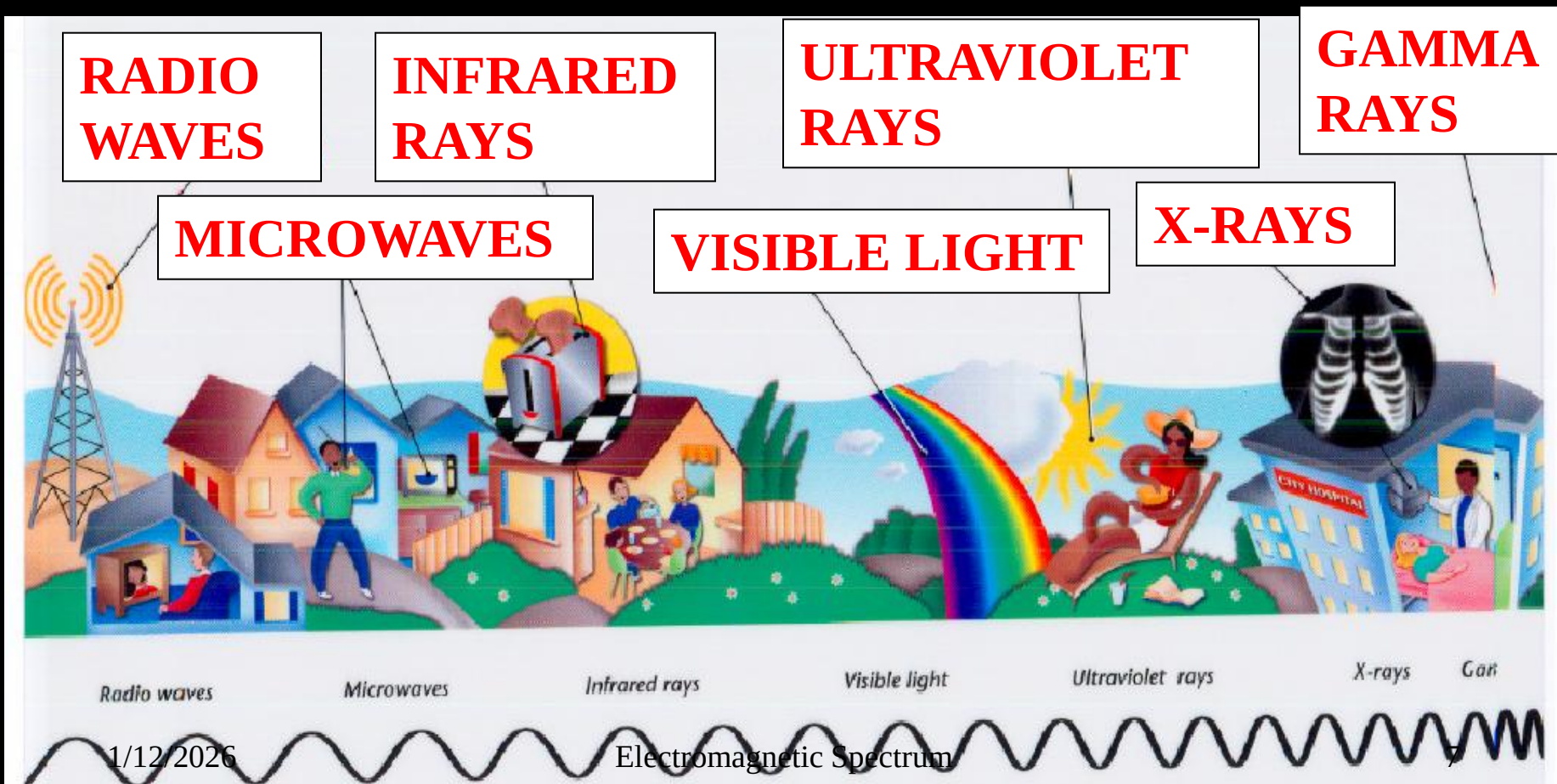


Waves or Particles?

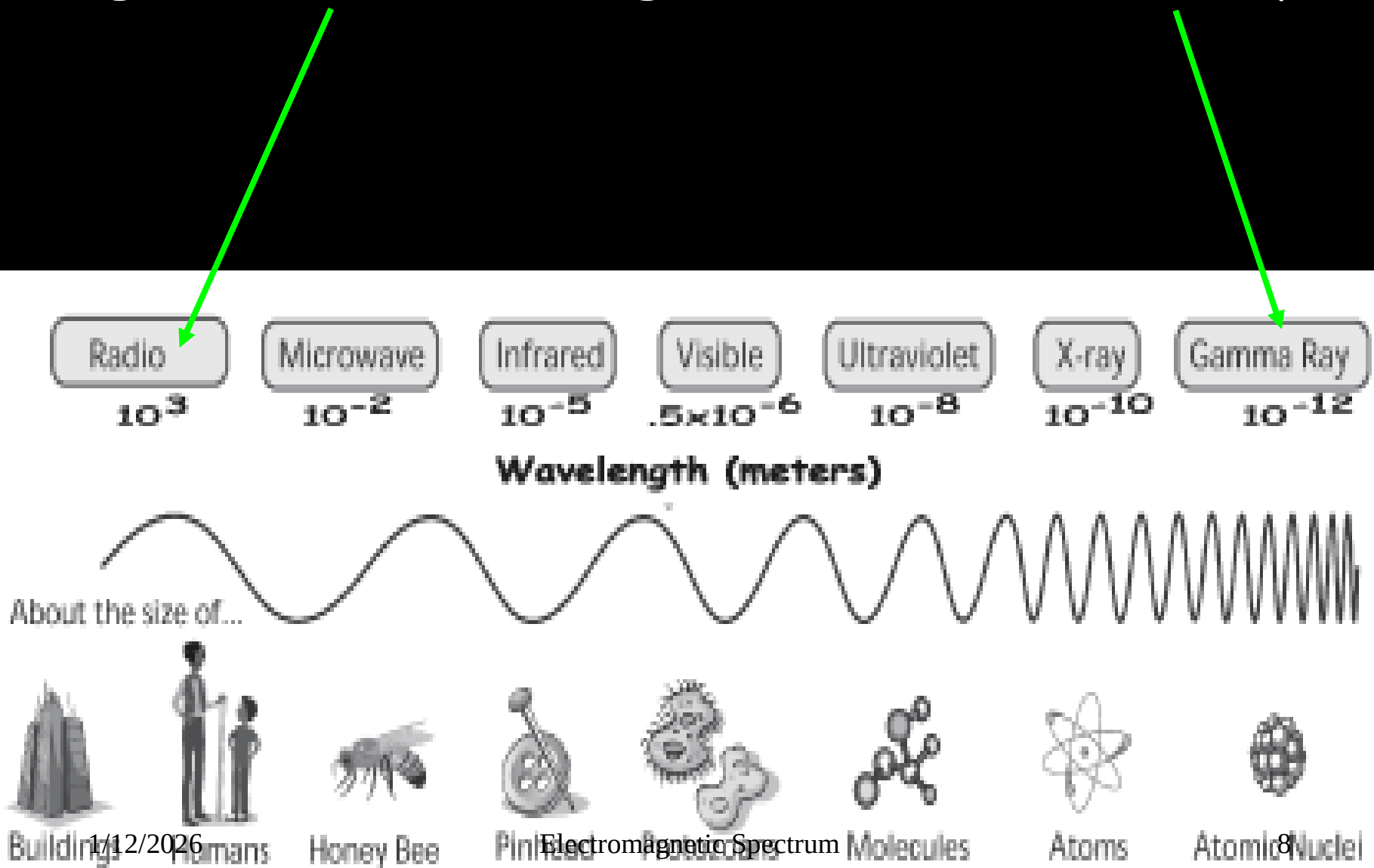
- Electromagnetic radiation has properties of waves but also can be thought of as a stream of particles.
- Example: Light
 - Light as a wave: Light behaves as a transverse wave which we can filter using polarized lenses.
 - Light as particles (photons): When directed at a substance light can knock electrons off of a substance (Photoelectric effect)



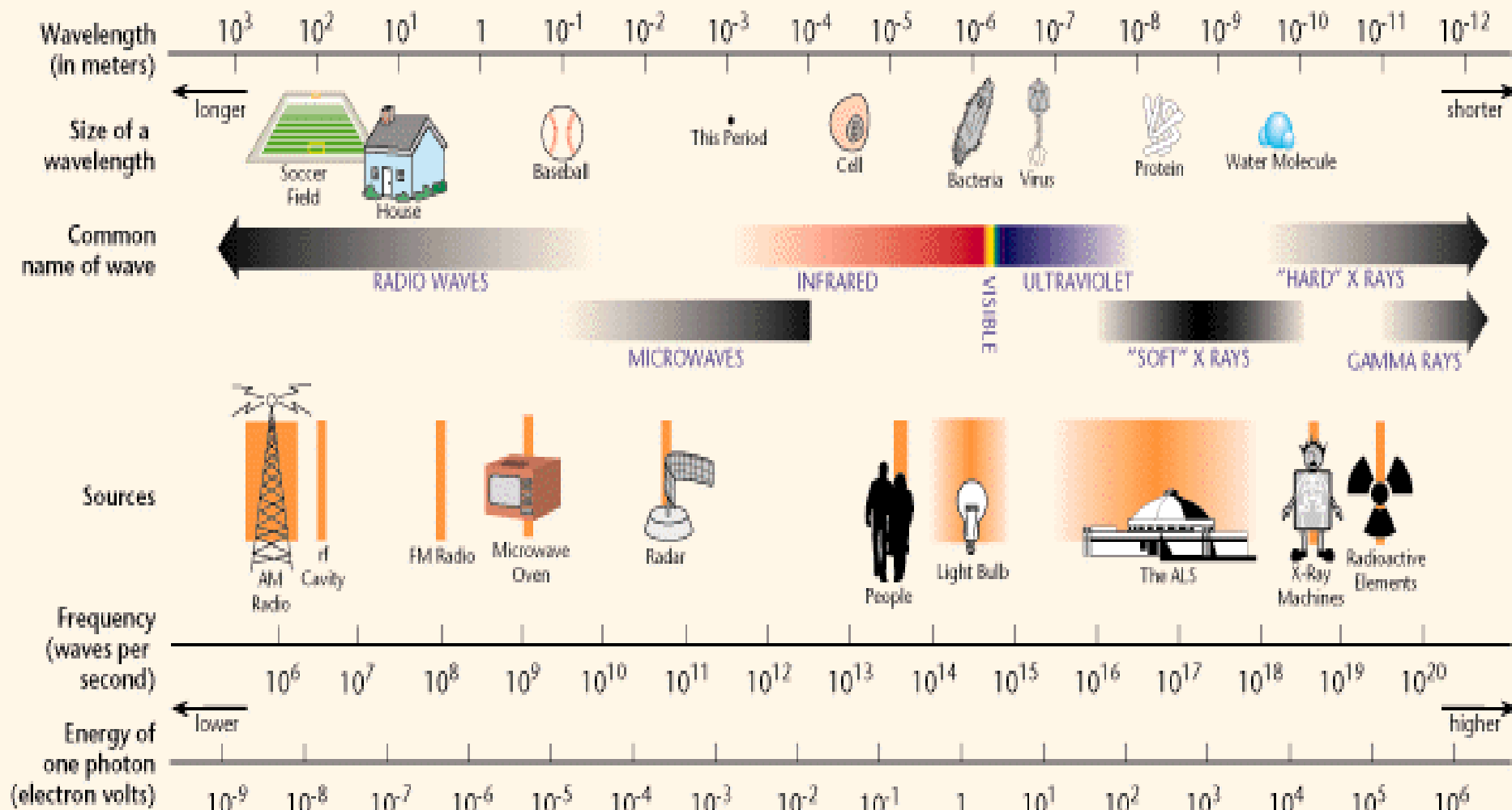
Electromagnetic Spectrum—name for the range of electromagnetic waves when placed in order of increasing frequency



Notice the wavelength is long (Radio waves) and gets shorter (Gamma Rays)

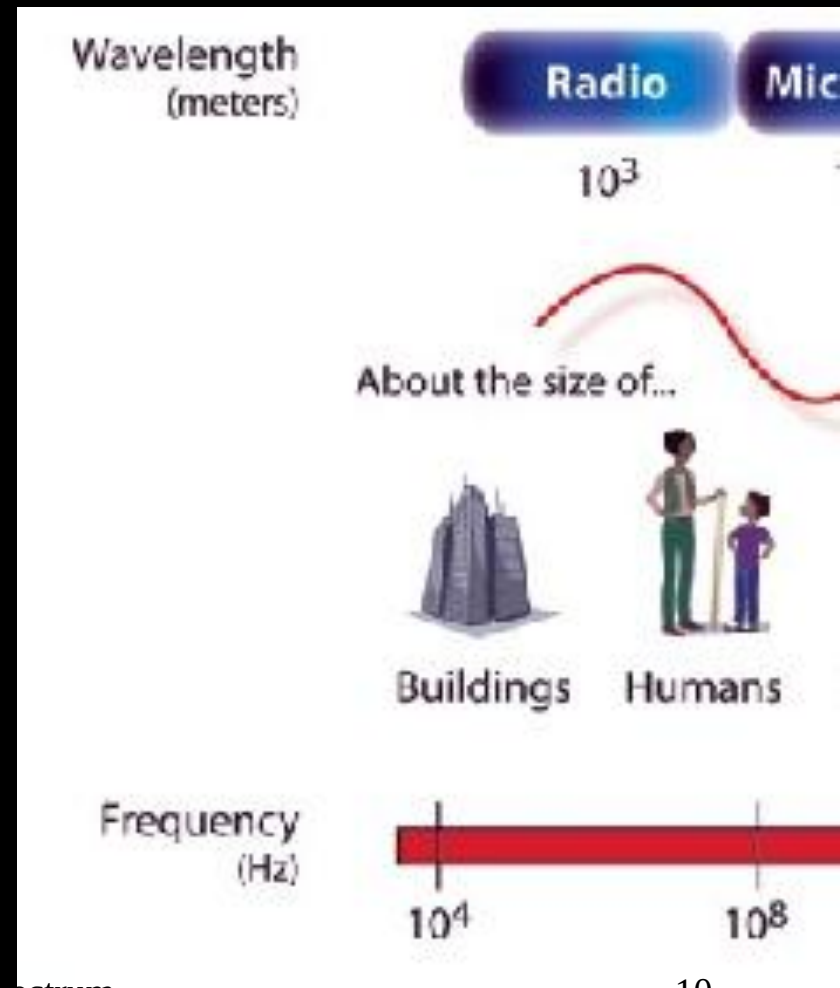


THE ELECTROMAGNETIC SPECTRUM

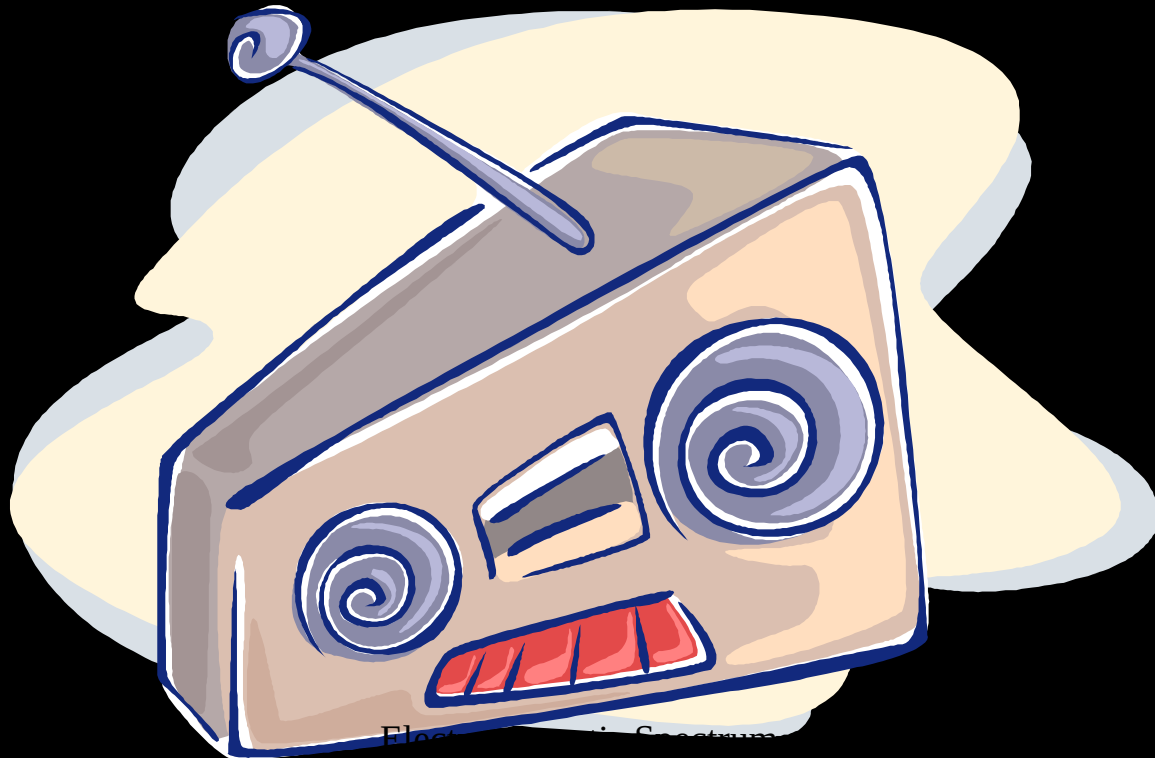


RADIO WAVES

Have the longest wavelengths and lowest frequencies of all the electromagnetic waves.



- A radio picks up radio waves through an antenna and converts it to sound waves.
- Each radio station in an area broadcasts at a different frequency.
 - # on radio dial tells frequency.



MRI

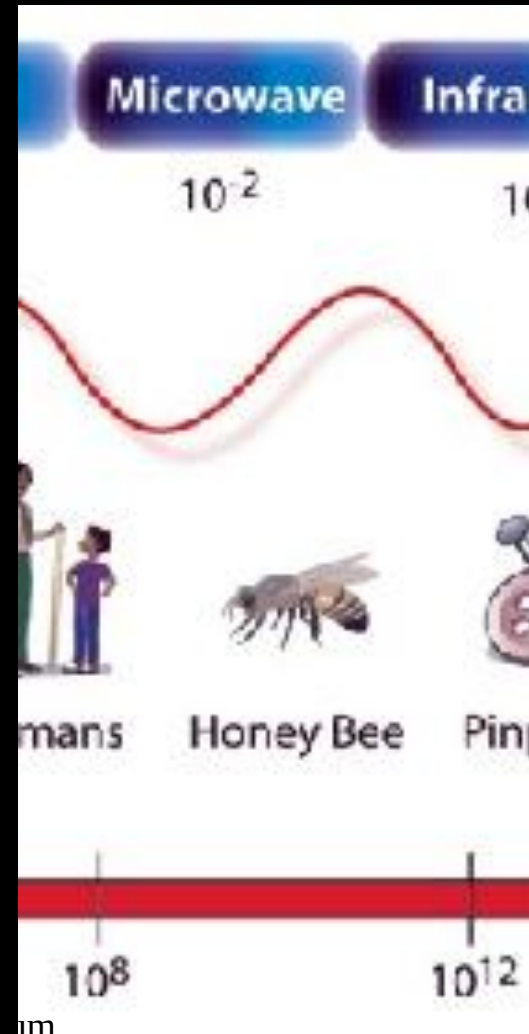
(MAGNETIC RESONANCE IMAGING)

Uses Short wave radio waves with a magnet to create an image.

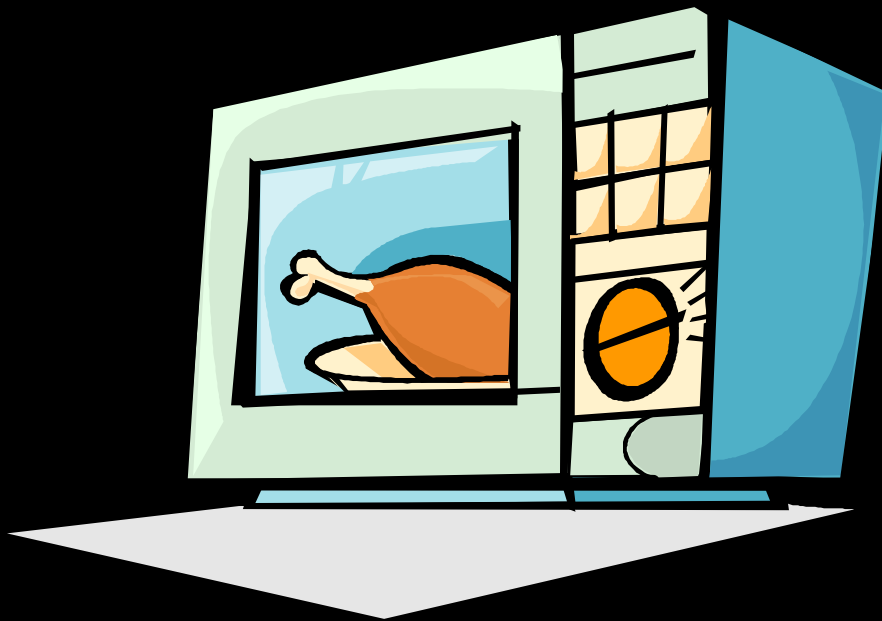


MICROWAVES

Have the
shortest
wavelengths and
the highest
frequency of
the radio
waves.



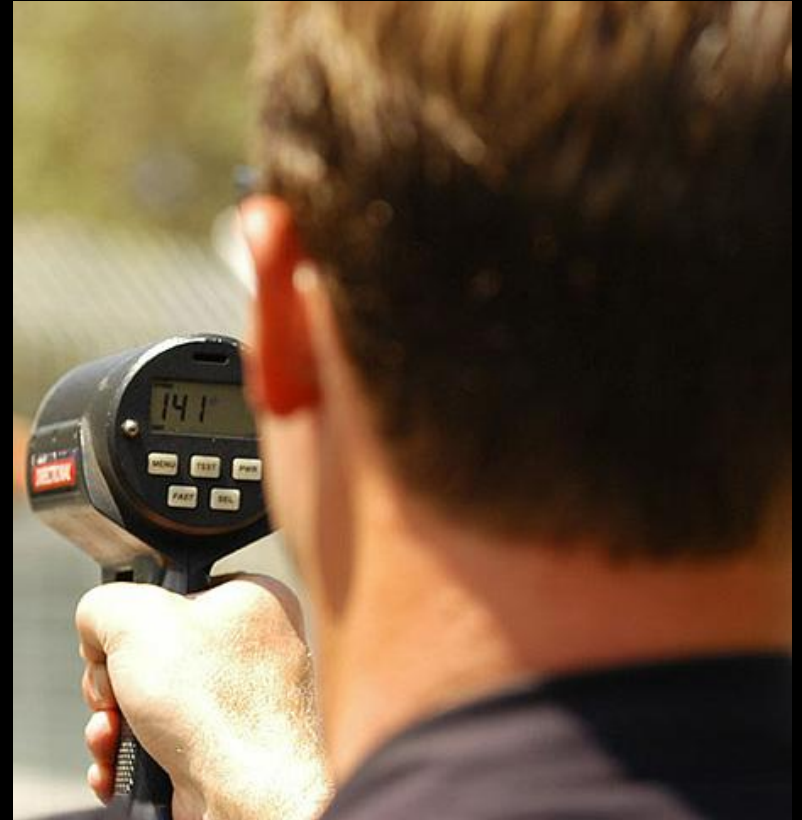
Used in microwave ovens.



- Waves transfer energy to the water in the food causing them to vibrate which in turn transfers energy in the form of heat to the food.

RADAR (Radio Detection and Ranging)

- Used to find the speed of an object by sending out radio waves and measuring the time it takes them to return.



INFRARED RAYS

Infrared = below
red

Shorter
wavelength and
higher
frequency than
microwaves.

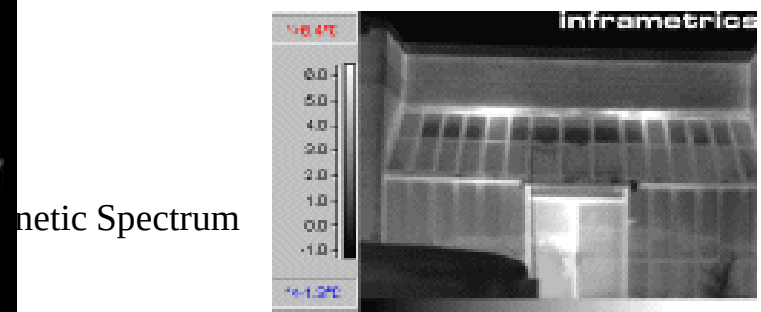
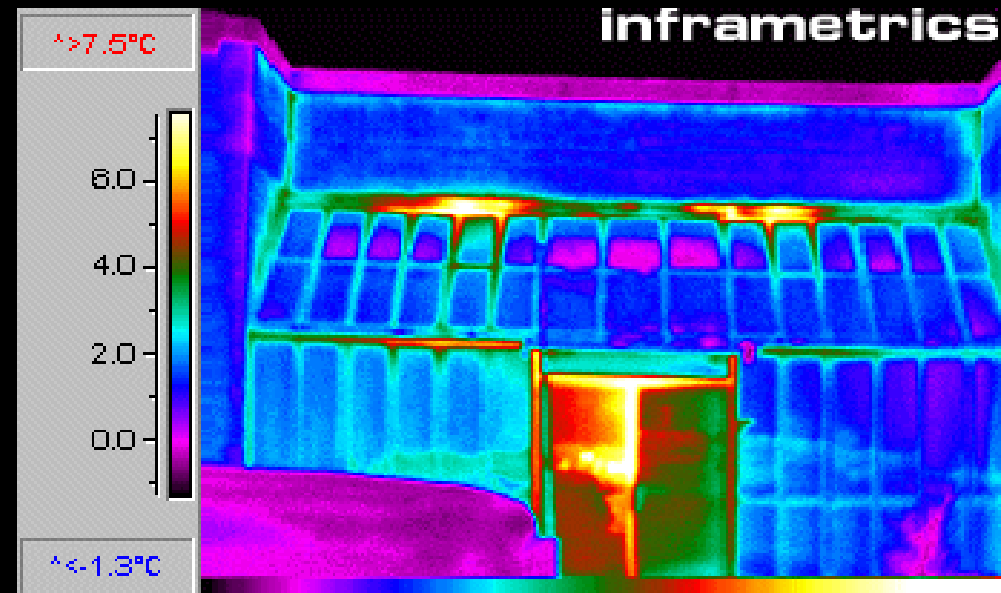


Thermogram—a picture that shows regions of different temperatures in the body. Temperatures are calculated by the amount of infrared radiation given off.

Therefore people give off infrared rays.



Heat lamps give off infrared waves.



Thermal Spectrum

VISIBLE LIGHT

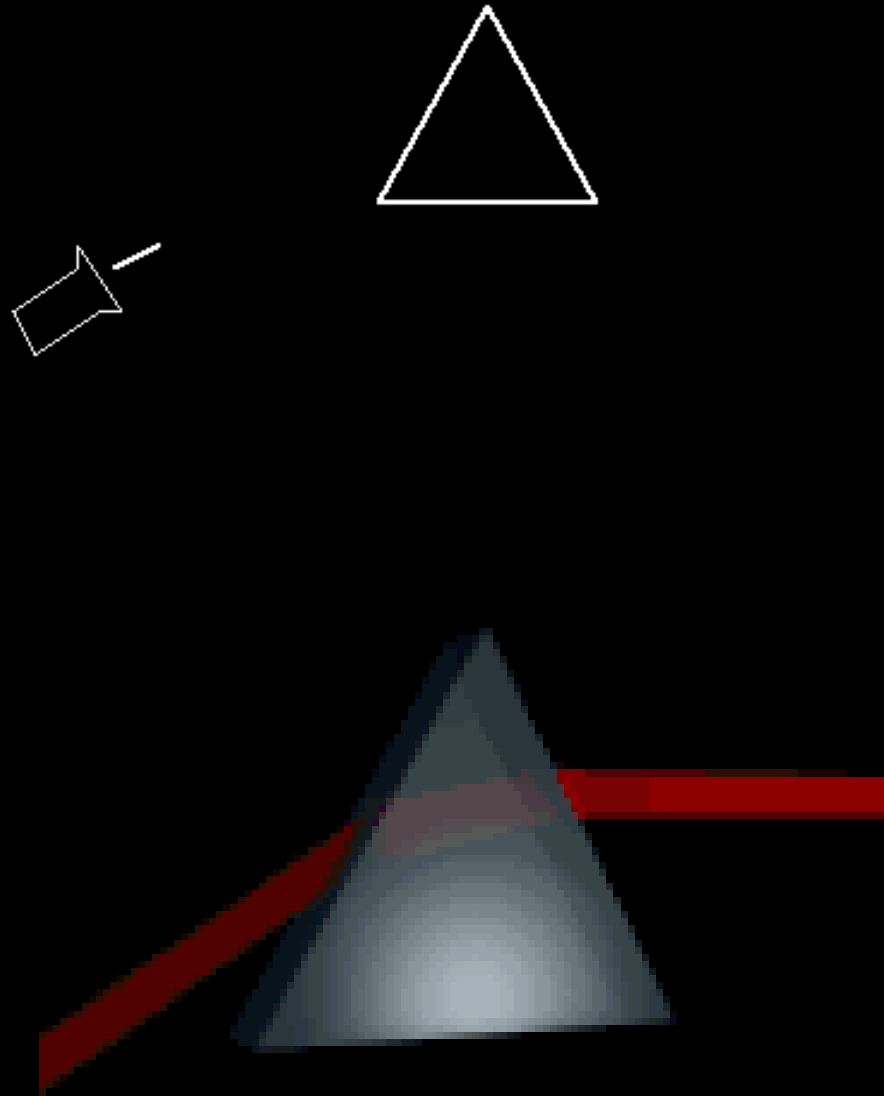
Shorter wavelength and
higher frequency than
infrared rays.

Electromagnetic waves we
can see.

Longest wavelength= red
light

Shortest wavelength=
violet (purple) light



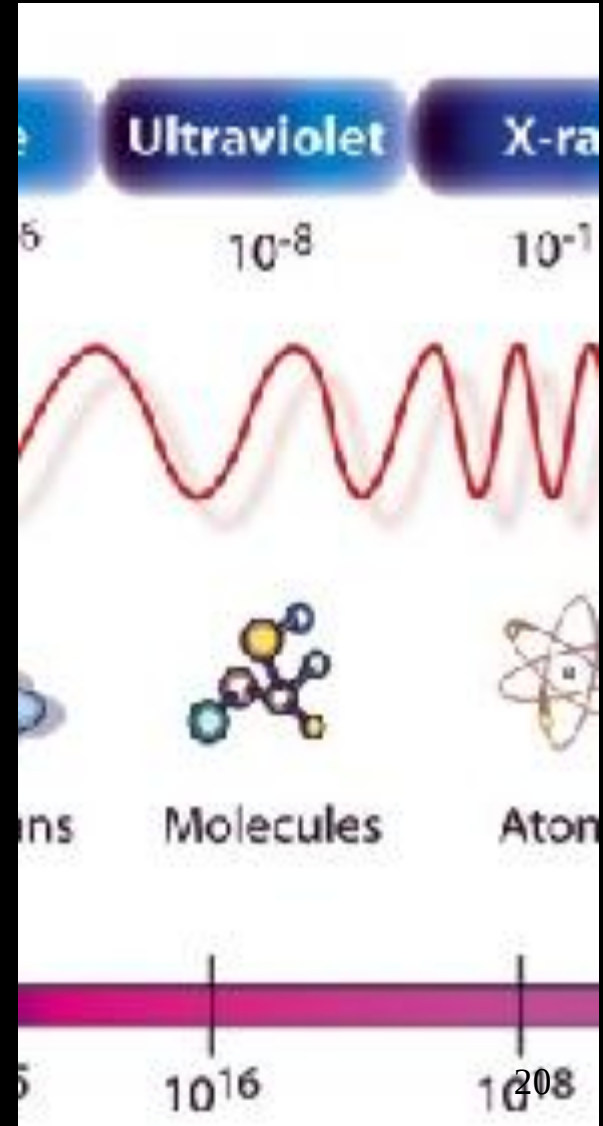


When light enters
a new medium it
bends (refracts).
Each wavelength
bends a different
amount allowing
white light to
separate into it's
various colors
ROYGBIV.

ULTRAVIOLET RAYS

Shorter
wavelength and
higher
frequency than
visible light

Carry more
energy than
visible light



Used to kill
bacteria.
(Sterilization
of equipment)



Too much can cause skin cancer.

Use sun block to protect against (UV rays)



1/12/2026

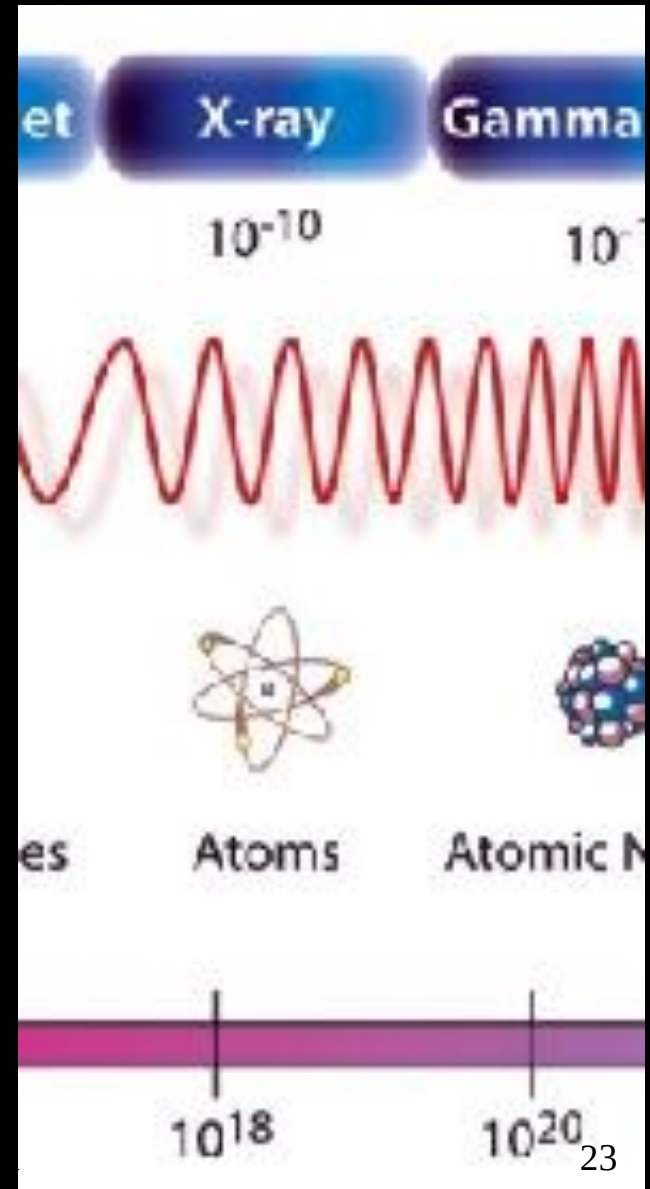
Electroma

X- RAYS

Shorter
wavelength and
higher
frequency than
UV-rays

Carry a great
amount of
energy

Can penetrate
most matter.

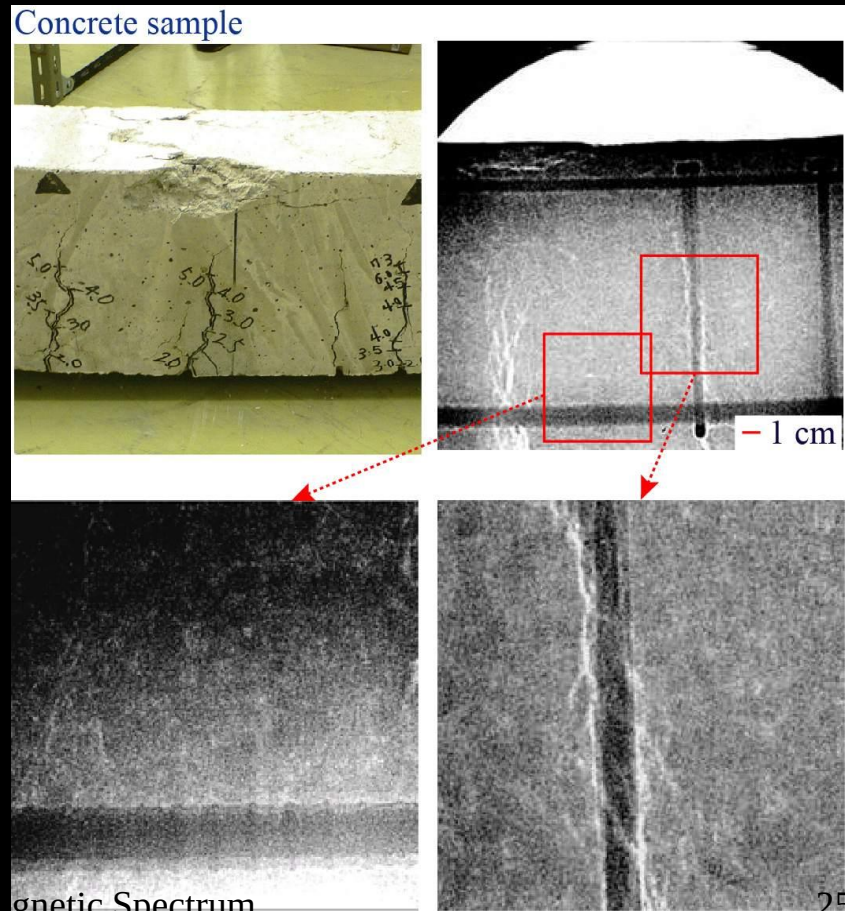


Bones and teeth absorb x-rays. (The light part of an x-ray image indicates a place where the x-ray was absorbed)



Used by engineers to check for tiny cracks in structures.

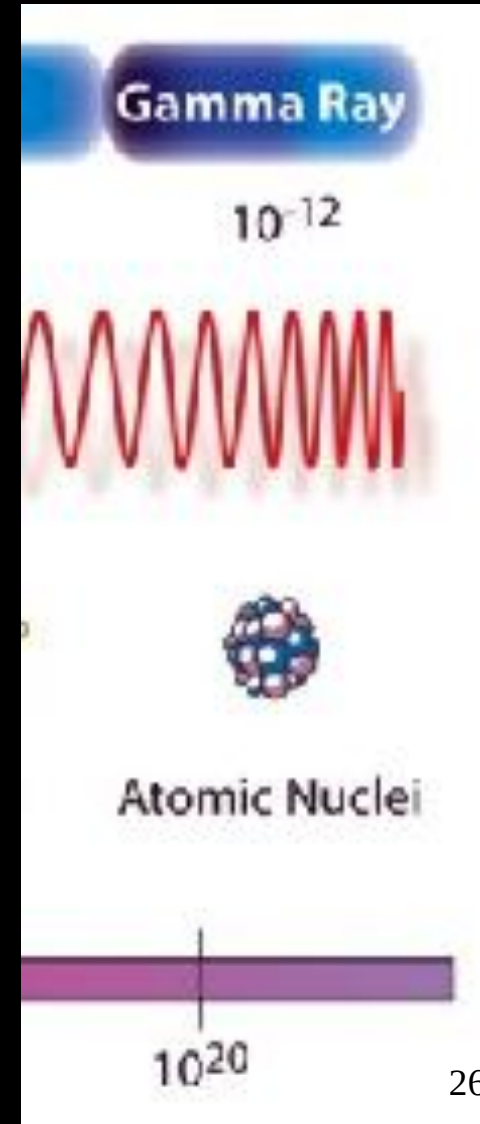
- The rays pass through the cracks and the cracks appear dark on film.



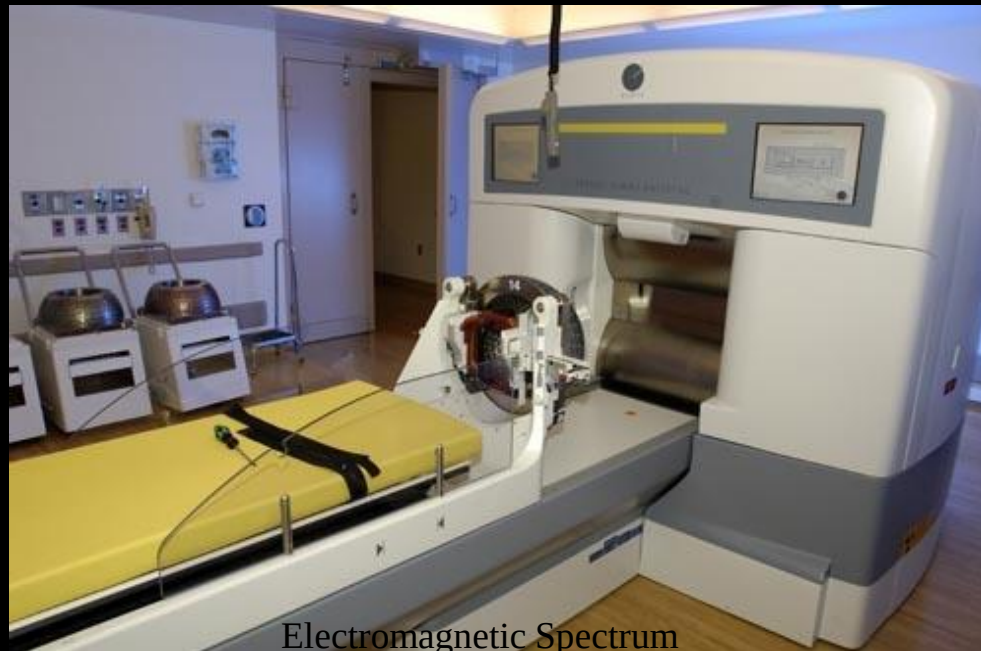
GAMMA RAYS

Shorter wavelength
and higher frequency
than X-rays

Carry the greatest
amount of energy
and penetrate the
most.



Used in radiation treatment to kill cancer cells.
Can be very harmful if not used correctly.



Electromagnetic Spectrum

Exploding
nuclear
weapons emit
gamma rays.



Brief SUMMARY

- A. All electromagnetic waves travel at the same speed. (300,000,000 meters/second) in a vacuum.
- B. They all have different wavelengths and different frequencies.
- Long wavelength → lowest frequency
 - Short wavelength → highest frequency
 - The higher the frequency the higher the energy.